

SPPD Flock Shared Network × ICE 287(g) Cross-Index Analysis

Methodology & Reproduction Guide

Data processed: May 2026
South Portland Police Department, Maine

1. Overview

This analysis cross-indexes the South Portland Police Department (SPPD) Flock Safety automatic license-plate reader (ALPR) shared-network roster against the U.S. Immigration and Customs Enforcement (ICE) official 287(g) participant list. The goal is to identify agencies that appear in **both** datasets — i.e., agencies that SPPD actively shares ALPR data with AND that hold an active 287(g) agreement with ICE, which deputizes local law enforcement to perform federal immigration enforcement functions.

The core question: **Is SPPD's ALPR data flowing to agencies with 287(g) agreements?** If so, license plate data collected in South Portland could be used to support federal immigration enforcement — even if SPPD itself has no such agreement.

The entire analysis has been reproduced in a standalone Excel workbook (sppd-287g-crossindex-methodology.xlsx) using basic formulas that require no Python. This document explains the methodology and provides step-by-step reproduction instructions.

Metric	Value
SPPD agencies sharing ALPR data	602
ICE 287(g) agencies (deduplicated)	1,503
Matched agencies (both datasets)	73
States with at least one match	27
Unique US states in SPPD sharing network	44
Automated matches (exact + Jaccard ≥ 0.75)	72
Manual matches (accepted)	1
Rejected false positives	2

Table 1. Key summary statistics

2. Data Sources

2.1 SPPD Shared Network

Obtained via a Freedom of Access Act (FOAA) request to the South Portland Police Department. The file `SharedNetworks_2026_May_5.csv` lists every agency in SPPD's Flock Safety network as of May 5, 2026, including the direction of sharing: which agencies SPPD shares **to** (`Networks I'm Sharing`) and which agencies share **to** SPPD (`Networks Shared With Me`).

Only rows where `Networks I'm Sharing` is non-null are included in this analysis — i.e., agencies SPPD is **actively sending data to**. Agencies that only send data to SPPD (one-way inbound) are excluded, as the concern is about data flowing **outward** from South Portland.

Source hash (SHA-256): 75729defe7be5ad5454423b815fba159bb456675813dec41fa1e7342f440c19c

2.2 ICE 287(g) Participant List

Downloaded from the ICE official website as a CSV with columns: state, agency, and support types (Jail Enforcement Model, Task Force Model, and/or Warrant Service Officer). The raw list was deduplicated to produce `ice-287g-deduped.csv` with 1,503 unique agency-state pairs.

287(g) is a federal program under § 287(g) of the Immigration and Nationality Act that allows ICE to enter into agreements with state and local law enforcement agencies, deputizing officers to perform immigration enforcement functions.

Source hash (SHA-256): aaf71a60705b1f8b907e2a6b1f2d35d1d49709154c5c25c147c8c2fc8110c9e9

2.3 Intermediate Normalized File

The Excel workbook ships with pre-normalized data. For full reproducibility from source, the intermediate `sppd-shared-networks-normalized.csv` is produced by a separate normalization step (documented in §3). Both input CSVs and this intermediate are embedded in the Excel workbook's raw data sheets.

3. Methodology

3.1 Step 1: State Extraction

Agency names in the SPPD spreadsheet often include a state abbreviation or full state name (e.g., Lake City FL PD or Massachusetts Department of Correction). The normalization process extracts the state and converts it to a canonical two-letter lowercase code (e.g., fl, ma).

Two-letter tokens that are NOT states — PD (Police Department), SO (Sheriff's Office), IB (Intake/Booking), LR (vendor tag) — are explicitly excluded.

The state is stored separately and used as a **hard constraint**: two agencies can only match if they are in the same state.

Example:

```
Lake City FL PD → state = fl
```

3.2 Step 2: State Suffix Removal

After extracting the state, all remaining state code tokens and full state names are stripped from the agency name in all positions (not just trailing).

3.3 Step 3: Agency Name Normalization

The agency name is cleaned through several transformations applied together:

- **Lowercase:** All text is lowercased for case-insensitive matching.
- **Suffix stripping:** Common agency-type suffixes are removed: Police Department, Police Dept, PD, Sheriff's Office, SO, Sheriff, County Sheriff, City, Town, Village, Borough, Township, Campus, College, University, System, Center, Network, District, Corrections, Jail, Prison, Facility, Board, Commission, Agency, Unit, Service(s), Authority, Department, Office, Division, Bureau, Command, Detectives, Garage, Booking, Intake, Constable, Precinct, Task Force, Warrant Service Officer, Jail Enforcement.
- **Punctuation removal:** All non-alphanumeric characters are removed.
- **Vendor/deployment tag removal:** Tokens like Flex, LR, Condor are removed.
- **IB preservation:** Intake/Booking (IB) is NOT stripped because it is a facility identifier, not an agency-type suffix. Only one SPPD entry has IB, and it does not match any 287(g) agency.
- **State code removal:** All two-letter state codes are removed from remaining tokens.
- **Collapsing:** Extra whitespace is collapsed to single spaces.

Example pipeline:

```
Henry County FL SO → extract state fl → strip state Henry County → suffix strip henry county → punctuation/whitespace henry county
```

3.4 Step 4: Pair Generation

For every SPPD agency and every ICE 287(g) agency that share the **same state code** (after normalization), a candidate pair is generated. This produces tens of thousands of pairs — all materialized in the Excel workbook's Pairs & Jaccard sheet.

When an SPPD agency has no extractable state code (e.g., CSX Railroad, MAGLOCLEN RISS), it is excluded from pair generation — no state-based fuzzy matching is possible, and exact-name matches across all states are not meaningful.

3.5 Step 5: Jaccard Similarity Scoring

For each pair, the Jaccard word-overlap similarity is computed. The Jaccard score measures how similar two sets of words are:

Jaccard(A, B) = $|A \cap B| \div |A \cup B|$ — the number of shared words divided by the total number of unique words across both names.

This is the **union denominator** formulation. A previous version used the minimum-size denominator (max intersection), which inflated scores on short names. The union formulation is the standard Jaccard definition and properly penalizes when one name contains many words not present in the other.

Worked example:

SPPD core words: fish wildlife commission (3 words)
ICE core words: florida fish wildlife conservation commission (5 words)
Intersection: {fish, wildlife, commission} (3 words)
Union: {fish, wildlife, commission, florida, conservation} (5 words)
Jaccard = $3/5 = 0.60$

3.6 Step 6: Matching Criteria

An SPPD agency matches an ICE agency if BOTH of these conditions hold:

- 1. Same state:** Both agencies have the same state code after normalization.
- 2. Name similarity:** Either the normalized core names are **exactly identical** (Jaccard = 1.0), OR the Jaccard score is ≥ 0.75 .

Stateless-match restriction:

When an SPPD agency has no extractable state, only exact normalized-name matches are permitted — no fuzzy Jaccard matches. This prevents short-name hits against ICE agencies in any random state.

The threshold of 0.75 was chosen empirically: it captures known variant spellings (correction vs. corrections) while rejecting false-positives like Pennsylvania State matching a local constable office. The Excel workbook allows changing this threshold in cell H2 on the Best Matches sheet.

3.7 Step 7: Best-Match Selection

When an SPPD agency generates multiple candidate ICE matches, the **single best match** is selected using this priority order:

- **Higher confidence score wins.**
- **Exact beats fuzzy:** Among equal scores, exact name matches are preferred over Jaccard ≥ 0.75 .
- **Shorter normalized name wins:** Among ties with the same match type, the ICE agency with the shorter (more specific) normalized name is preferred.

This selection happens automatically via the MAXIFS and INDEX/MATCH formulas in the Excel workbook.

3.8 Step 8: Manual Mappings

Some agencies are known matches but fall below the automated Jaccard threshold. These are recorded as manual mappings:

SPPD Name	ICE 287(g) Agency	Jaccard	Verdict	Reason
Fish and Wildlife Commission FL PD	Florida Fish & Wildlife Conservation Commission	0.60	ACCEPTED	SPPD omits 'Florida' and 'Conservation'. Below 0.75 but clearly the same agency.
Pennsylvania State PD	Pennsylvania State Constable's Office, Cooke Township	—	REJECTED	PA State Police is statewide; Cooke Township constable is local. Short-prefix false positive.
Bradford County Detectives PA	Bradford County Sheriff's Office (PA)	—	REJECTED	Detectives is a sub-unit within the SO, not the SO itself. Different entities.

Table 2. Manual mappings

3.9 Confidence Scoring

Score	Tier	Meaning
1.0	exact	Normalized core names identical after stripping
0.75–0.99	fuzzy	Jaccard word-overlap ≥ 0.75 (union denominator)
< 0.75	manual	Accepted via manual mapping; automated score below threshold

Table 3. Confidence scoring tiers

3.10 Duplicate Flag

When multiple SPPD agencies map to the **same ICE agency** in the same state, a `duplicate_flag` is set. This is informational — it signals vendor/deployment variants in the SPPD spreadsheet (e.g., Flex and non-Flex entries for the same real-world agency). It does not invalidate the match.

4. Reproducing the Analysis in Excel

The companion Excel workbook (`sppd-287g-crossindex-methodology.xlsx`) reproduces the entire analysis pipeline using simple formulas. No Python, macros, or external tools are required. Every intermediate value can be inspected, every formula can be traced, and every threshold can be modified.

4.1 Workbook Structure

The workbook contains 9 sheets:

Sheet	Role	Formulas?
Read Me	Methodology overview and formula inventory	No
SPPD Normalized	602 SPPD agencies with state codes	No (data)
ICE 287(g)	1,503 ICE agencies with core names	No (data)
Pairs & Jaccard	30,514 same-state pairs with Jaccard scores	No (pre-computed)
Best Matches	Best ICE match per SPPD agency	Yes — live threshold
Manual Mappings	Hand-validated accepted/rejected pairs	No
Final Results	73 matched agencies	Yes — COUNTIF summaries
State Counts	44 US states with agency counts	Yes — SUM, COUNTIF
Jaccard Demo	5 step-by-step intersection/union walkthroughs	No

Table 4. Excel workbook sheets

4.2 Key Formulas Used

The workbook uses only functions available in Microsoft Excel 2013+ and LibreOffice Calc:

- **SUM:** Total agency counts across states.
- **COUNTIF / COUNTIFS:** Count matches per condition (by method, by state, duplicate flag, ICE 287(g) presence per state).
- **IF:** Threshold checks — determines whether a Jaccard score meets the threshold (≥ 0.75) and classifies matches as exact, fuzzy, or none.
- **MAXIFS:** (Excel 2019+) Finds the highest Jaccard score for an SPPD-agency/state combination from the Pairs sheet. Alternative: manually sort and filter.
- **INDEX + MATCH:** (Excel 2019+) Returns the ICE agency name for the best match. Alternative: manual lookup in the Pairs sheet.

4.3 Reproduction Steps

Follow these steps to verify or reproduce any finding manually:

Step A — Verify a specific match:

1. Open the Final Results sheet. Pick any row (e.g., henry county al matched to Henry County Sheriff's Office).
2. Note the ICE state and normalized core name. Go to the ICE 287(g) sheet, filter by that state, and verify the agency exists.
3. Compare the normalized core name column in ICE with the SPPD core name (shown in Pairs & Jaccard). For exact matches, they will be identical after stripping suffixes and state codes.
4. For fuzzy matches (score between 0.75 and 0.99), count the words in common and the total unique words. Divide common by unique.

Step B — Change the Jaccard threshold:

1. Open the Best Matches sheet.
2. Locate cell H2 (current value: 0.75).
3. Change it to 0.80, 0.85, or 1.0 and watch column I update: rows showing YES meet the threshold, No do not.
4. At 0.80: Massachusetts Department of Correction drops out (Jaccard 0.75).
5. At 1.00: only exact name matches remain.

Step C — Verify the state count:

1. Open the State Counts sheet.
2. Column A lists every state where SPPD has partner agencies. Count rows (excluding '(no state)') — the result is 44.
3. Column D marks which states have ICE 287(g) agencies (36 of 44). The COUNTIF formula at the bottom computes this.
4. Note: 8 states in SPPD's network (DE, CA, etc.) have no ICE 287(g) agencies registered. Matches can only occur in the other 36.

Step D — Rebuild from raw sources:

1. Start with the raw FOAA file: SharedNetworks_2026_May_5.csv.
2. Filter to rows where Networks I'm Sharing is non-blank.
3. Extract the state code from each agency name (manual lookup against US state list).
4. Strip the state code, suffixes, punctuation, and vendor tags to get the core name.
5. For each SPPD agency, find all ICE agencies in the same state and compute the Jaccard score (count common words ÷ count unique words across both).
6. Apply the 0.75 threshold and best-match selection. Add manual mappings. Count unique states.

5. Results

5.1 Matched Agencies by Confidence Tier

Of the 8 matched pairs, the majority are exact matches:

- **Exact matches (1.0):** 72 matched agencies. The normalized core name is identical after stripping suffixes and state codes — e.g., `brevard county` exactly matches `brevard county`.
- **Fuzzy matches (≥ 0.75):** 0 agencies. All fuzzy scores below 1.0 either meet the threshold for exact or are captured manually.
- **Manual match:** 1 agency — Fish and Wildlife Commission FL PD → Florida Fish & Wildlife Conservation Commission (Jaccard 0.60).

5.2 Top States by Match Count

State	Matches	% of Total
FL (Florida)	22	30.1%
TX (Texas)	6	8.2%
PA (Pennsylvania)	4	5.5%
NC (North Carolina)	3	4.1%
MO (Missouri)	3	4.1%
KS (Kansas)	3	4.1%
OK (Oklahoma)	3	4.1%
AL, AR, AZ, GA, ID, KY, LA, MA, MI, MS, NE, NH, NV, NY, OH, SC, SD, TN, VA, WV	1–2 each	$\leq 3\%$ each

Table 5. Top states for SPPD-ICE 287(g) matches

5.3 Support Types Among Matched Agencies

The 287(g) program has three models of participation. Matched agencies span all three:

- **Task Force Model:** Officers perform immigration enforcement during routine patrol duties. Most common among matches.
- **Jail Enforcement Model:** Officers identify and process removable noncitizens in jails. Present in county-level agencies.
- **Warrant Service Officer:** Officers serve administrative warrants on noncitizens. Often combined with other models.
- Many agencies participate in multiple models simultaneously (e.g., Jail Enforcement Model; Task Force Model; Warrant Service Officer).

5.4 State Network Reach

SPPD shares Flock ALPR data with agencies in **44 of 50 U.S. states** (plus DC). This is a genuinely national surveillance network, extending far beyond Maine. Texas, Florida, North Carolina, Ohio, Tennessee, and Arizona are among the most represented states.

6. Known Limitations

Naming convention variability. The SPPD shared-network spreadsheet uses varying naming conventions (e.g., 'High Springs FL PD Flex' and 'High Springs FL PD' for the same agency). Name normalization addresses this but may miss some variants.

ICE official list limitations. The ICE 287(g) list uses full legal names; SPPD uses shorthand. The Jaccard threshold of 0.75 is deliberately conservative to minimize false positives, but may miss legitimate variant-name matches.

No independent verification. 287(g) status has NOT been independently verified beyond comparing names against the ICE official list. Agencies on the list may have terminated or suspended agreements not yet reflected.

Agencies not on the ICE list. Agencies that share Flock data with SPPD but are NOT on the ICE list may still have 287(g) agreements not reflected in the official list, or may have de facto coordination with ICE outside the 287(g) framework.

Sub-unit matching. Sub-units within large agencies (e.g., 'Bradford County Detectives') may still match their parent agency after suffix stripping. Manual rejection handles known cases; others may exist.

Ambiguous common names. Agencies with common names (e.g., 'Springfield') could match ICE agencies in unintended states. The state-constrained matching prevents cross-state errors, but within-state ambiguity remains.

One-directional data flow context. Agencies SPPD does NOT share with (receive-only and two-way-without-outbound) are excluded. Findings should be reported as 'among the {n} agencies SPPD actively shares data with' to avoid overstatement.

No geolocation matching. This analysis uses name matching only. Geographic proximity, phone numbers, addresses, and organizational hierarchies are not used.

7. Citation & Provenance

To cite this analysis (APA-style):

No Flock For SoPo. (2026, May). *SPPD Flock Shared Network × ICE 287(g) Cross-Index Analysis: Methodology & Reproduction Guide*. Retrieved from [audit-log-analysis-202605/outputs/excel-methodology/](#)

Primary Source Files:

- `SharedNetworks_2026_May_5.csv` — SPPD FOAA response (May 5, 2026).
- `ice-287g-deduped.csv` — ICE official 287(g) participant list, deduplicated from ICE.gov.

Derived Outputs:

- `crossindex_results.csv` — Python-generated cross-index (73 rows).
- `sppd-287g-crossindex-methodology.xlsx` — Excel methodology workbook.
- `sppd-287g-crossindex-methodology.pdf` — This document.

All files and source data are version-controlled and committed to the `audit-log-analysis-202605` repository. File hashes are provided for auditability.

End of document. For questions or errata, contact the project maintainer.